

1.1 Antacids

Antacids are chemical substances that increase the pH of gastric contents thereby neutralizing the acidic contents of stomach, thus providing symptomatic relief in ulcer dyspepsia and non-erosive gastro-oesophageal reflux. Liquid preparations are more effective than solid preparations.

ALUMINIUM HYDROXIDE

Dosage form and strength: *Oral suspension:* Aluminium hydroxide 250 mg/5 ml and Magnesium hydroxide 250 mg/5 ml; *Tablets/Gel:* may contain simethicone, and some contain algenic acid

Indication: Ulcer and non ulcer dyspepsia; gastro-oesophageal reflux.

Contraindication/Precautions: Hypophosphatemia, undiagnosed GI or rectal bleeding, appendicitis, porphyria, renal impairment (aluminium may accumulate), constipation, hepatic impairment, dehydration.

Dosage schedule:

- *Tablets:* 1-2 tablets chewed 4 times a day and at bed time or as required;
- *Suspension:* 5-10 ml 4 times daily between meals and at bedtime; Children (6-12 years): up to 5 ml 3 times daily

Adverse effects: Constipation, haemorrhoids, fissures, faecal impaction

Drug and food interaction: Decreases absorption of digoxin, azithromycin, ciprofloxacin, norfloxacin, rifampicin, tetracycline, isoniazid

Patient information: Take with water to decrease constipation

MAGNESIUM HYDROXIDE

Dosage form and strength: *See under aluminium hydroxide*

Indication: Ulcer and non ulcer dyspepsia, gastro-oesophageal reflux

Contraindication/Precautions: Severe renal impairment, renal impairment, hepatic impairment.

Dosage schedule: 250-500 mg, per oral 3-4 times a day

Adverse effects: Diarrhoea in renal impairment, respiratory depression, hypotension, loss of tendon reflexes, cardiac arrest

Drug and food interaction: Decrease absorption of other drugs

Patient information: Do not take this medicine within 2 hours of taking other medicines (especially flouroquinolones); causes diarrhoea

1.2 Antispasmodics

ATROPINE SULPHATE

Dosage form and strength: *Injection:* 0.6 mg/ml

Indication: Gastrointestinal spasm, renal colic, anesthetic premedication, OP poisoning

Contraindication/Precautions: Avoid in case of angle closure glaucoma, hypersensitivity, tachycardia secondary to cardiac insufficiency or thyrotoxicosis. Use in pregnancy only when potential benefits justify possible risk to fetus.

Dosage schedule:

- *Premedication:* intravenous injection, 200-600 micrograms immediately before induction of anesthesia and in incremental doses of 100 micrograms for the treatment of bradycardia; intramuscular injection, 300-600 micrograms 30-60 minutes before induction; child 20 micrograms/kg;
- *For control of muscarinic side effects of neostigmine:* in reversal of competitive neuromuscular block, by intravenous injection, 0.6-1.2 mg;
- *Bradycardia:* particularly complicated by hypotension after myocardial infarction, intravenous injection of 300 micrograms, increasing to 1 mg if necessary;
- *Antidote to organophosphorous poisoning:* by intramuscular or intravenous injection, 1 to 2 mg, repeated in 20-30 minutes as soon as cyanosis has cleared

Adverse effects: Dry mouth, constipation, urinary retention, blurred vision

Drug and food interaction: Anticholinergic effect is more pronounced with tricyclic antidepressants, amantidine, antiparkinsonism agents; decreases the absorption of ketoconazole, levodopa; antacids decrease the effect of atropine

Patient information: Do not perform strenuous activity in high temperatures, heat stroke may occur; do not perform tasks requiring motor co-ordination and alertness (e.g. driving)

DROTAVERINE

Dosage form and strength: *Tablet:* 10 mg, 40 mg and 80 mg; *Injection:* 20 mg/ml

Indication: Gastrointestinal colicky pain, renal colic, 1st stage of labour (dilation of cervical opening)

Contraindication/Precautions: Paralytic ileus (or about to undergo surgery), gall bladder disease, diabetes, depression, ulcerative colitis, heart disease, liver disease, kidney disease

Dosage schedule: 40-80 mg 3 times per day

Adverse effects: Nausea, vomiting, fainting, dry mouth

FLAVOXATE HYDROCHLORIDE

Dosage form and strength: *Tablet:* 100 mg and 200 mg

Indication: Increased urinary frequency and incontinence, bladder spasms due to catheterization, urgency, dysuria

Contraindication/Precautions: Obstructive uropathy, ileus, GI obstruction, GI bleeding, glaucoma

Dosage schedule: 200 mg 3 times daily

Adverse effects: *Same as in hyoscine*, fatigue, vertigo

Drug and food interaction: Can be taken without regard to food

Patient information: Do not perform tasks requiring motor co-ordination/alertness (e.g. driving)

HYOSCINE BUTYL BROMIDE

Dosage form and strength: *Tablet:* 10 mg and 20 mg; *Injection:* 20 mg/ml

Indication: Symptomatic GI and pain to urinary spasm, irritable bowel syndrome, excessive respiratory secretion, bowel colic

Contraindication/Precautions: Pregnancy, geriatric patient, hyperthyroidism, dysarrhythmias, ulcerative colitis, hypertension, renal disease, hepatic disease, urinary retention

Dosage schedule: 10 mg 3 times a day; 4 to 20 mg 4 times a day

Adverse effects: Headache, confusion, dizziness, hallucination, palpitation, tachycardia, blurred vision, tachycardia, blurred vision, photophobia, cycloplegia, dry mouth, constipation, paralytic ileus

MEBEVERINE HYDROCHLORIDE

Dosage form and strength: *Tablet:* 135 mg; *Suspension:* 50 mg/5ml

Indication: Irritable bowel syndrome, other spasmodic conditions

Contraindication/Precautions: Paralytic ileus. To be taken 20 minutes before meal

Dosage schedule: 135-150 mg 3 times a day to be taken 20 minutes before meal

Adverse effects: Allergic reaction, angioedema, urticaria, rash

1.3 Ulcer healing drugs

CIMETIDINE

Dosage form and strength: *Tablets:* 200 mg and 400 mg; *Injection:* 100 mg/ml

Indication: *see Ranitidine*

Contraindication/Precautions: *see Ranitidine*

Dosage schedule:

- *Benign ulcer (oral):* 400 mg twice daily followed by 800 mg once at night for 4 weeks, maintenance 400 mg at night; *Child:* 25-30 mg/kg; *Infant:* 20 mg/kg in divided doses;
- *Injection(IM):* 200 mg every 4-6 hrs, *Injection(IM):* 200 mg slowly over 2 minutes; *IV infusion:* 400 mg in 100 ml of normal saline over 1½-1 hours, 50-100 mg/hour over 24 hrs for continuous infusion

Adverse effects: *See ranitidine*, gynaecomastia, microsomal enzyme inhibition and hepatotoxicity are higher

Drug and food interaction: *See ranitidine*

ESOMEPRAZOLE

Dosage form and strength: *Tablets:* 20 mg and 40 mg; *Injection:* 40 ml/vial

Indication: *See omeprazole*

Contraindication/Precautions: *See omeprazole*

Precautions: *See omeprazole*

Dosage schedule:

- *Duodenal ulcer associated with H. pylori:* 20 mg twice daily
- *NSAID associated GU:* 20mg once daily for 4-8 weeks; for prophylaxis with increase rate of gastroduodenal complication who require continued NSAID treatment 20 mg daily

• *GERD*: 40 mg once daily for 4 weeks, continued for further 4 weeks if not fully healed or symptom persist, maintenance 20 mg daily

Adverse effects: *See omeprazole*

Drug and food interaction: *See omeprazole*

FAMOTIDINE

Dosage form and strength: *Tablets: 20 mg and 40mg; Injection: 10 mg/ml*

Indication: *See ranitidine*

Contraindication/Precautions: *See ranitidine*

Dosage schedule:

- *Benign ulcer*: Oral 40 mg at night for 4-8 weeks, maintenance dose 20 mg at night
- *Reflux oesophagitis*: 20-40 mg twice daily for 5-12 weeks
- *Zollinger–Ellison syndrome*: 20 mg every 6 hours

Adverse effects: *See ranitidine*

Drug and food interaction: *See ranitidine*

Patient information: Avoid alcohol, NSAIDs, extreme hot and spicy foods

LANSOPRAZOLE

Dosage form and strength: *Capsule: 15 mg and 30 mg*

Indication: *See omeprazole*

Contraindication/Precautions: *See omeprazole*

Dosage schedule:

- *Duodenal ulcer*: 30 mg daily in morning for 4 week, maintenance 15 mg daily; *Benign gastric ulcer*: 30 mg daily in morning for 8 week;
- *NSAID associated duodenal or gastric ulcer*: 15-30 mg once daily for 4 weeks continued for further 4 weeks if not fully healed;
- *Reflux oesophagitis*: 30 mg daily in morning for 4 weeks, continued for further 4 weeks if not fully healed;
- *Zollinger–Ellison syndrome*: initially 60 mg once daily adjusted according to response, daily dose of 120 mg or more given in two divided doses;
- *Reflux oesophagitis refractory to other treatment*: 40 mg for 8 weeks

Adverse effects: *See omeprazole*

Drug and food interaction: *See omeprazole*

OMEPRAZOLE

Dosage form and strength: *Capsule: 10 and 20mg*

Indication: Benign gastric and duodenal ulcer, NSAID associated duodenal or gastric ulcer, duodenal or benign gastric ulcer associated with *Helicobacter pylori*, reflux oesophagitis, Zollinger–Ellison syndrome, stress ulcer

Contraindication/Precautions: Hypersensitivity reaction, breast-feeding, patients with liver disease

Dosage schedule: Adult: 20 mg/day for 4-8 week

Adverse effects: Nausea, vomiting, diarrhoea, abdominal colic, skin rash, headache and dizziness

Drug and food interaction: Inhibits hepatic metabolism of diazepam, phenytoin, flurazepam, digoxin etc; increases the bleeding tendency with